

First, we start by adding this file to Splunk. It is not a typical Windows log, so I used the regex <Event> to divide the events.

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: 4688_xml_windows_security_delete_shadow.log

View Event Summary

Source type: Select Source Type

Save As

Event Breaks

Define event boundaries for incoming data.

Event-breaking policy

AutoEvery lineRegex

Pattern

<Event>

* Specifies a regex that determines how the raw text stream is broken into initial events, before line merging takes place.
* This sets SHOULD_LINEMERGE = false and LINE_BREAKER to the user-provided regular expression.
* Defaults to (\n|), meaning data is broken into an event for each line, delimited by any number of carriage return or newline characters.
* The regex must contain a capturing group -- a pair of parentheses which defines an identified subcomponent of the match.
* Wherever the regex matches, Splunk considers the start of the first capturing group to be the end of the previous event, and considers the end of the first capturing group to be the start of the next event.
* The contents of the first capturing group are discarded, and will not be present in any event. You are telling Splunk that this text comes between lines.

FormatSelect...Select...

	Time	Event
1	7/8/26 9:50:40.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-10-03T15:53:58.73556200Z' /><EventRecordID>202090</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='6036' /><Channel>Security</Channel><Computer>ar-win-dc.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>Administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0x11d0a1c</Data><Data Name='NewProcessId'>0x1acc</Data><Data Name='NewProcessName'>C:\Windows\System32\wbem\WMIC.exe</Data><Data Name='TokenElevationType'>%X1936</Data><Data Name='ProcessId'>0x950</Data><Data Name='CommandLine'>wmic.exe shadowcopy delete</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserSid'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event> timestamp = none
2	7/8/26 9:50:40.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-10-03T15:53:58.695119200Z' /><EventRecordID>202088</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='6036' /><Channel>Security</Channel><Computer>ar-win-dc.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>Administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0x11d0a1c</Data><Data Name='NewProcessId'>0x950</Data><Data Name='NewProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='TokenElevationType'>%X1936</Data><Data Name='ProcessId'>0x180</Data><Data Name='CommandLine'>cmd.exe /c "wmic.exe shadowcopy delete"</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserSid'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event>

```
source="4688_xml_windows_security_delete_shadow.log" sourcetype="wins"
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| search EventID=4688
```

We will be searching for Process Creation events. Event ID 4688 indicates that a new process was created.

source="4688_xml_windows_security_delete_shadow.log" host="si-i-0ca3967f682e00a13.prd-p-tjo7h.splunkcloud.com" sourcetype="wins"

Time range: All time

Search

4 events (before 7/8/26 10:45:56.000 PM) No Event Sampling

Job

Policy-Based Pool

Smart Mode

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect 1 millisecond per column

Hide FieldsAll Fields

SELECTED FIELDS
host 1
source 1
sourcetype 1

INTERESTING FIELDS
EventID 1
Guid 1
Index 1
linecount 1
Name 1
ProcessID 1
punct 1
splunk_server 1
SystemTime 4
ThreadID 2
timestamp 1
xmlns 1

+ Extract New Fields

FormatShow: 20 Per PageView: List

i	Time	Event
>	7/8/26 9:51:22.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-10-03T15:53:57.968409900Z' /><EventRecordID>202085</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='3008' /><Channel>Security</Channel><Computer>ar-win-dc.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>Administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0x11d0a1c</Data><Data Name='NewProcessId'>0x1b6c</Data><Data Name='NewProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='TokenElevationType'>%X1936</Data><Data Name='ProcessId'>0x180</Data><Data Name='CommandLine'>vssadmin.exe delete shadows /all /quiet</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserSid'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event> host = si-i-0ca3967f682e00a13.prd-p-tjo7h.splunkcloud.com source = 4688_xml_windows_security_delete_shadow.log sourcetype = wins
>	7/8/26 9:51:22.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-10-03T15:53:57.998961700Z' /><EventRecordID>202087</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='3008' /><Channel>Security</Channel><Computer>ar-win-dc.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>Administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0x11d0a1c</Data><Data Name='NewProcessId'>0x1b6c</Data><Data Name='NewProcessName'>C:\Windows\System32\vssadmin.exe</Data><Data Name='TokenElevationType'>%X1936</Data><Data Name='ProcessId'>0x1b6c</Data><Data Name='CommandLine'>vssadmin.exe delete shadows /all /quiet</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserSid'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event>

Even though I only have 4 events in this dataset, for the sake of practice I will write the queries as if I were investigating 40,000 events.

```
source="4688_xml_windows_security_delete_shadow.log" host="" sourcetype="wins"
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| rex field=_raw "<Data Name='NewProcessName'>(?!<NewProcessName>.*?)</Data>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>.*?)</Data>"
| search EventID=4688
| table NewProcessName CommandLine
```

This query extracts the New Process Name and Command Line from every Event ID 4688 event and displays them in a table.

Common commands:

vssadmin
wmic
wbadmin
bcdedit
diskshadow
powershell

Common keywords:

delete
shadow
shadowcopy
shadows
backup
catalog
recovery
recoveryenabled
bootstatuspolicy

I would then use the following query:

```
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>.*?)</Data>"
| search EventID=4688
| search CommandLine="*vssadmin*" OR CommandLine="*wmic*" OR CommandLine="*wbadmin*" OR
CommandLine="*bcdedit*" OR CommandLine="diskshadow"
```

First IOC

Now I have found the first IOC.

i	Time	Event																																																
▼	7/8/26 9:51:22.000 PM	<pre><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-10-03T15:53:57.968409900Z' /><EventRecordID>202085</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='3008' /><Channel>Security</Channel><Computer>ar-win-dc.attackrange.local</Computer><Security/></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>Administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0x11d0a1c</Data><Data Name='NewProcessId'>0x16bc</Data><Data Name='NewProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='TokenElevationType'>%X1936</Data><Data Name='ProcessId'>0x180c</Data><Data Name='CommandLine'>'cmd.exe' /c "vssadmin.exe delete shadows /all /quiet"</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserName'></Data><Data Name='TargetDomainName'></Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\Windows Explorer</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event></pre> Add to search Exclude from search Event Actions ▼ New search																																																
		<table border="1"> <thead> <tr> <th>Type</th><th>Field</th><th>Value</th><th>Actions</th></tr> </thead> <tbody> <tr> <td>Selected</td><td>host ▼</td><td>si-h-ca3967f682e00a13.prd-p-tjo7h.splunkcloud.com</td><td>▼</td></tr> <tr> <td></td><td>source ▼</td><td>4688_xml_windows_security_delete_shadow.log</td><td>▼</td></tr> <tr> <td></td><td>sourcetype ▼</td><td>wins</td><td>▼</td></tr> <tr> <td>Event</td><td>CommandLine ▼</td><td>"cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"</td><td>▼</td></tr> <tr> <td></td><td>EventID ▼</td><td>4688</td><td>▼</td></tr> <tr> <td></td><td>Guid ▼</td><td>'{54849625-5478-4994-a5ba-3e3b0328c30d}'</td><td>▼</td></tr> <tr> <td></td><td>Name ▼</td><td>'Microsoft-Windows-Security-Auditing'</td><td>▼</td></tr> <tr> <td></td><td>ProcessID ▼</td><td>'4'</td><td>▼</td></tr> <tr> <td></td><td>SystemTime ▼</td><td>'2023-10-03T15:53:57.968409900Z'</td><td>▼</td></tr> <tr> <td></td><td>ThreadID ▼</td><td>'3008'</td><td>▼</td></tr> <tr> <td></td><td>timestamp ▼</td><td>none</td><td>▼</td></tr> </tbody> </table>	Type	Field	Value	Actions	Selected	host ▼	si-h-ca3967f682e00a13.prd-p-tjo7h.splunkcloud.com	▼		source ▼	4688_xml_windows_security_delete_shadow.log	▼		sourcetype ▼	wins	▼	Event	CommandLine ▼	"cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"	▼		EventID ▼	4688	▼		Guid ▼	'{54849625-5478-4994-a5ba-3e3b0328c30d}'	▼		Name ▼	'Microsoft-Windows-Security-Auditing'	▼		ProcessID ▼	'4'	▼		SystemTime ▼	'2023-10-03T15:53:57.968409900Z'	▼		ThreadID ▼	'3008'	▼		timestamp ▼	none	▼
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	ThreadID ▼	'3008'	▼																																															
	timestamp ▼	none	▼																																															

Now here found the first IOC →

What we know that:

User: is **ATTACKRANGE\Administrator**. It is possible the attacker had already obtained administrator privileges.

Process: cmd.exe

Parent Process: powershell.exe

Command: vssadmin.exe delete shadows /all /quiet

vssadmin.exe — Runs the Windows Volume Shadow Copy management tool.

delete shadows — Deletes shadow copies (snapshots of files and disks).

/all — Deletes all shadow copies.

/quiet — Does it without asking for confirmation.

Ransomware often runs this command before encrypting files so victims cannot restore their files from shadow copies.

Second IOC

Second IOC which happened less than one second later

```
> 7/8/26 9:51:22.000 PM <Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4688</EventID><Version>2</Version><Level>0</Level><Task>13312</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime='2023-10-03T15:53:58.695119200Z' /><EventRecordID>202088</EventRecordID><Correlation><Execution ProcessID='4' ThreadID='6036' /><Channel>Security</Channel><Computer>ar-win-dc.attackrange.local</Computer><Security></System><EventData><Data Name='SubjectUserSid'>ATTACKRANGE\Administrator</Data><Data Name='SubjectUserName'>Administrator</Data><Data Name='SubjectDomainName'>ATTACKRANGE</Data><Data Name='SubjectLogonId'>0x11d0a1c</Data><Data Name='NewProcessId'>0x950</Data><Data Name='NewProcessName'>C:\Windows\System32\cmd.exe</Data><Data Name='TokenElevationType'>%%1936</Data><Data Name='ProcessId'>0x180</Data><Data Name='CommandLine'>"cmd.exe" /c "wmic.exe shadowcopy delete"</Data><Data Name='TargetUserSid'>NULL SID</Data><Data Name='TargetUserName'>-</Data><Data Name='TargetDomainName'>-</Data><Data Name='TargetLogonId'>0x0</Data><Data Name='ParentProcessName'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='MandatoryLabel'>Mandatory Label\High Mandatory Level</Data></EventData></Event>
```

host = si-i-0ca3967f682e00a13.prd-p-tjo7h.splunkcloud.com | source = 4688_xml_windows_security_delete_shadow.log | sourcetype = wins

User: ATTACKRANGE\Administrator

Process: cmd.exe

Parent Process: powershell.exe

Command Executed: wmic.exe shadowcopy delete

This command has the same goal as the first one. It attempts to delete the Volume Shadow Copies, but instead of using vssadmin, it uses Windows Management Instrumentation (WMI).

I think anything related to delete, shadow, or shadowcopy would likely be detected by security rules.

Conclusion and possible containment

As a Tier 1 SOC Analyst, **if permitted** by my organization's procedures, I would terminate the malicious process chain involving powershell.exe, cmd.exe, and the recovery deletion commands (vssadmin.exe and wmic.exe) if the attack is still active. I would preserve the logs and evidence, then escalate the incident to the Tier 2 SOC team for further investigation. I would also determine whether the Administrator account was legitimately used or if it had been compromised before taking action against the account. Or any other action done by the user ATTACKRANGE\Administrator